

# DISTRIBUCIÓN DE ESPECIES Y CAMBIO CLIMÁTICO

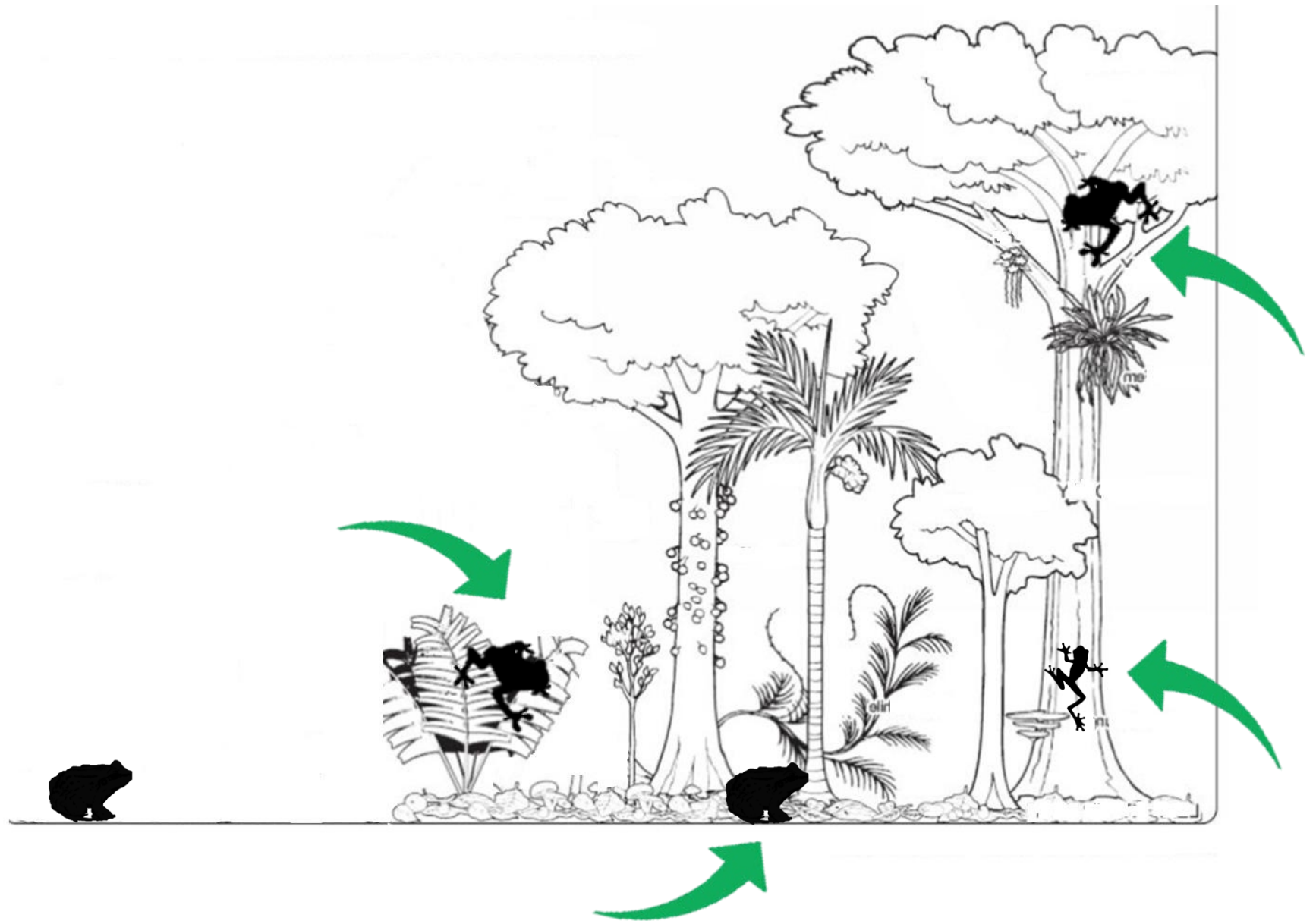
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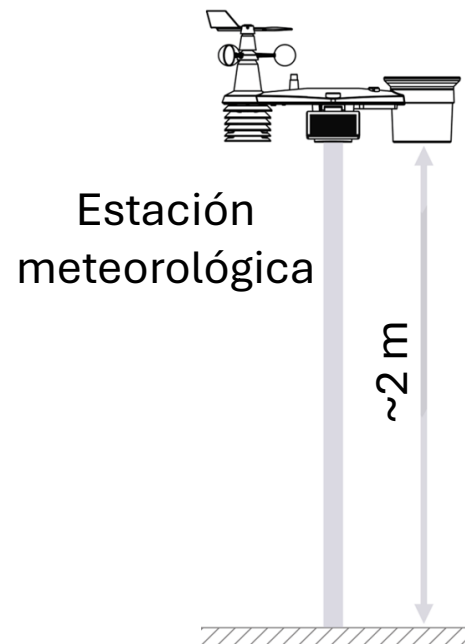
## Día 4 – Microambiente

Luis Miguel Senzano



São Paulo, Brasil.



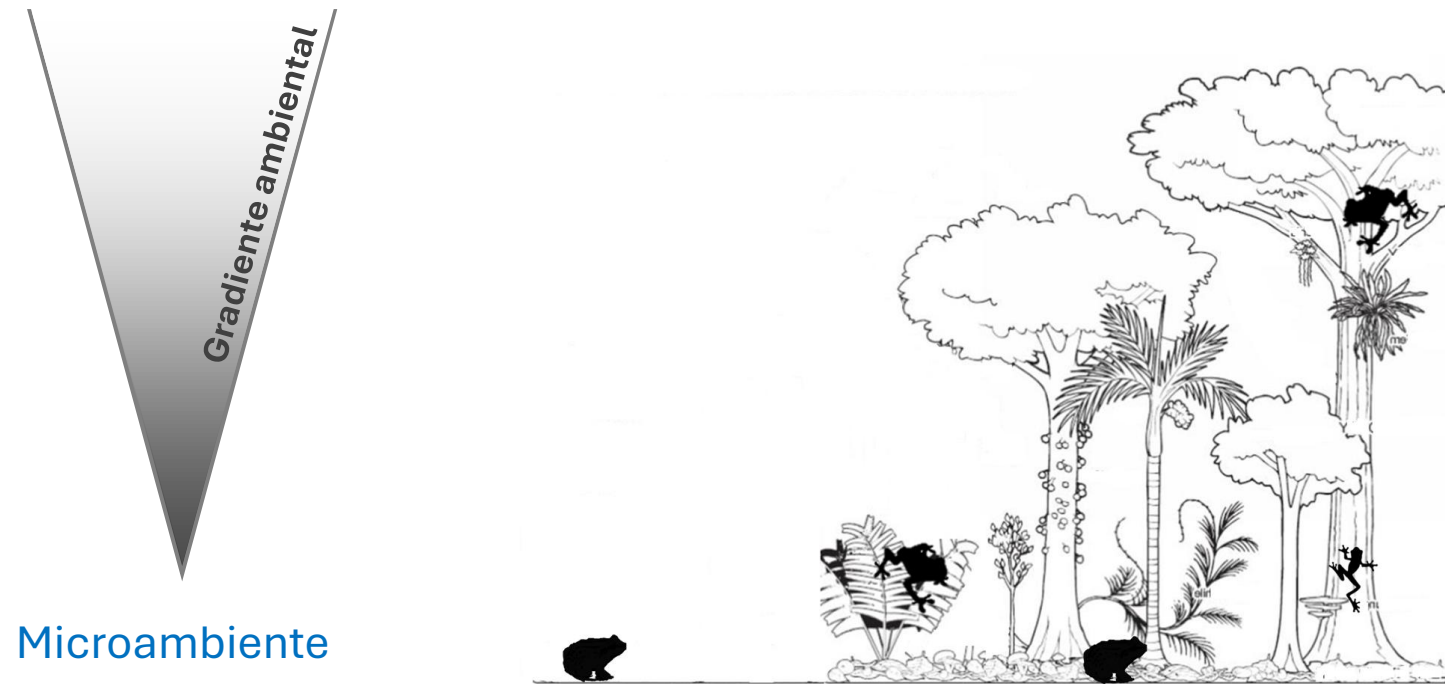


Condiciones ambientales representadas



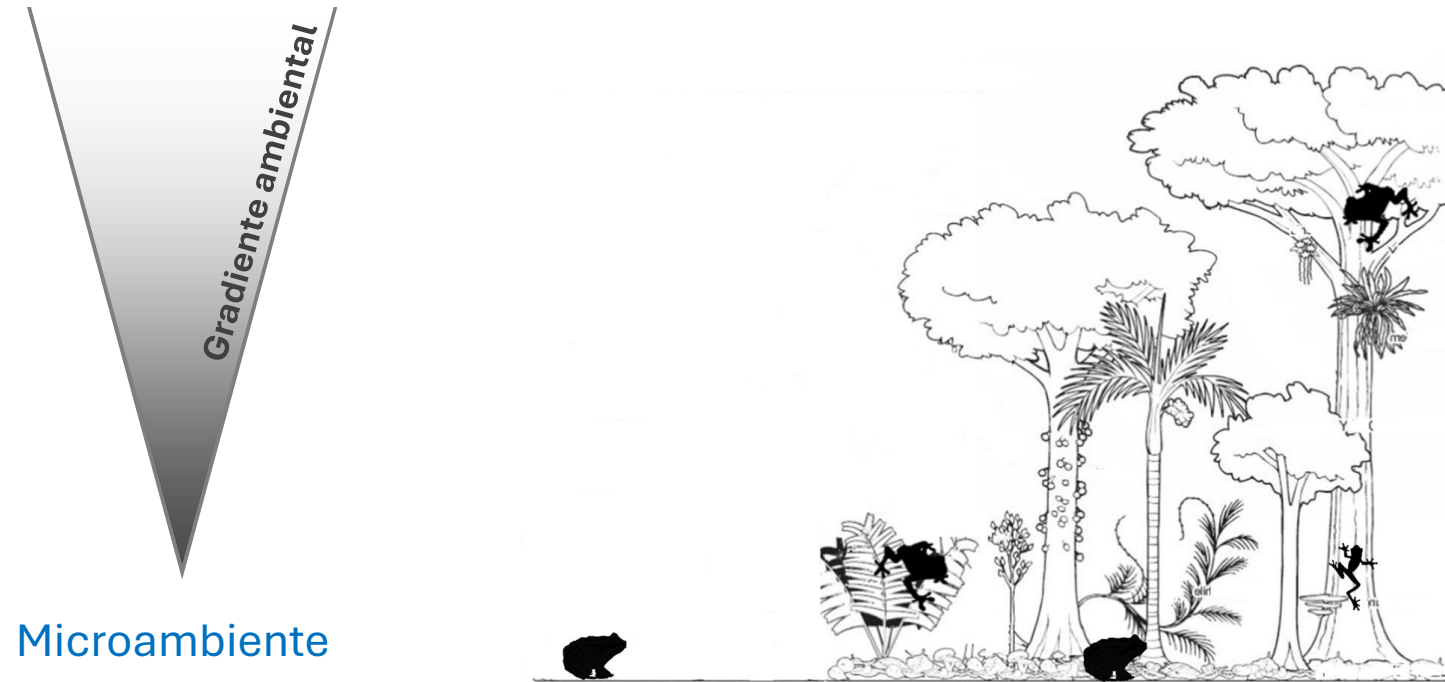
# MODELOS MACROCLIMÁTICOS

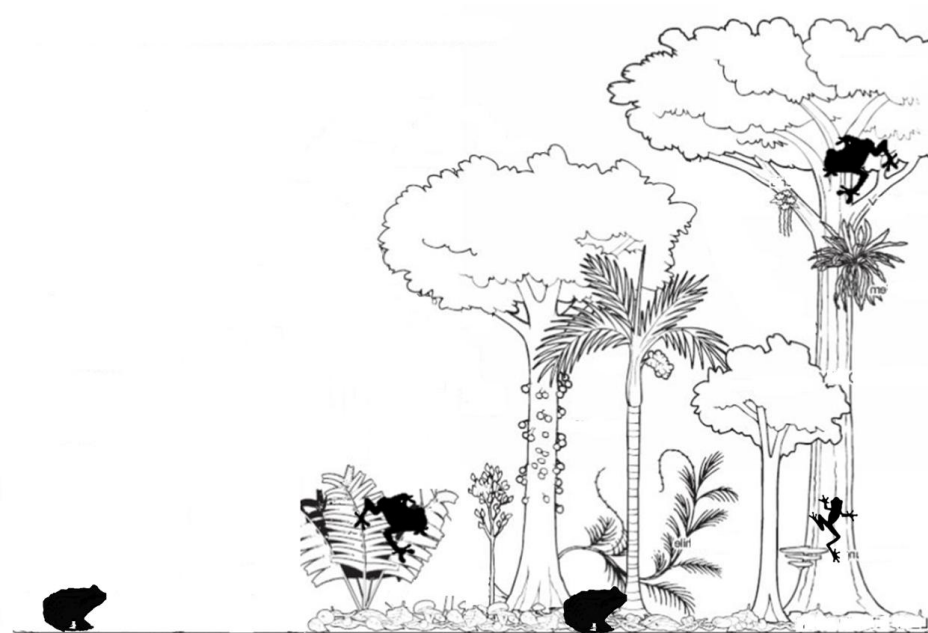
(e.g., WorldClim, ERA5, CHELSA, CMIP6, etc.)



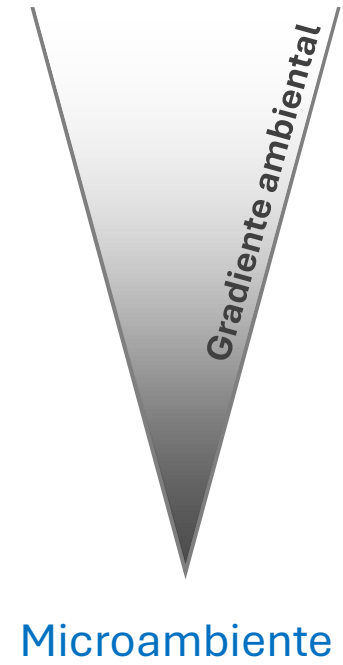
# MODELOS MACROCLIMÁTICOS

(e.g., WorldClim, ERA5, CHELSA, CMIP6, etc.)





## MODELOS MACROCLIMÁTICOS





Nivel de sombra

0%

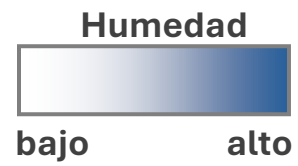
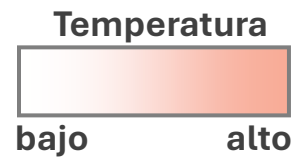
100%



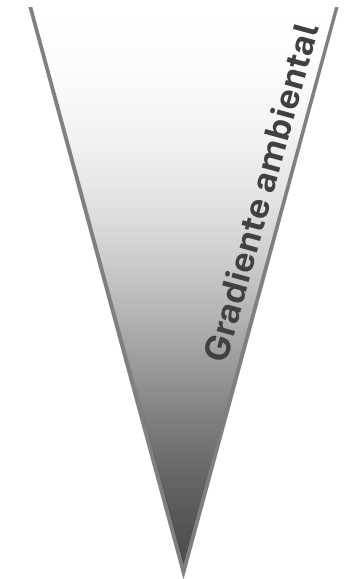
MODELOS MACROCLIMÁTICOS

Gradiente ambiental

Microambiente



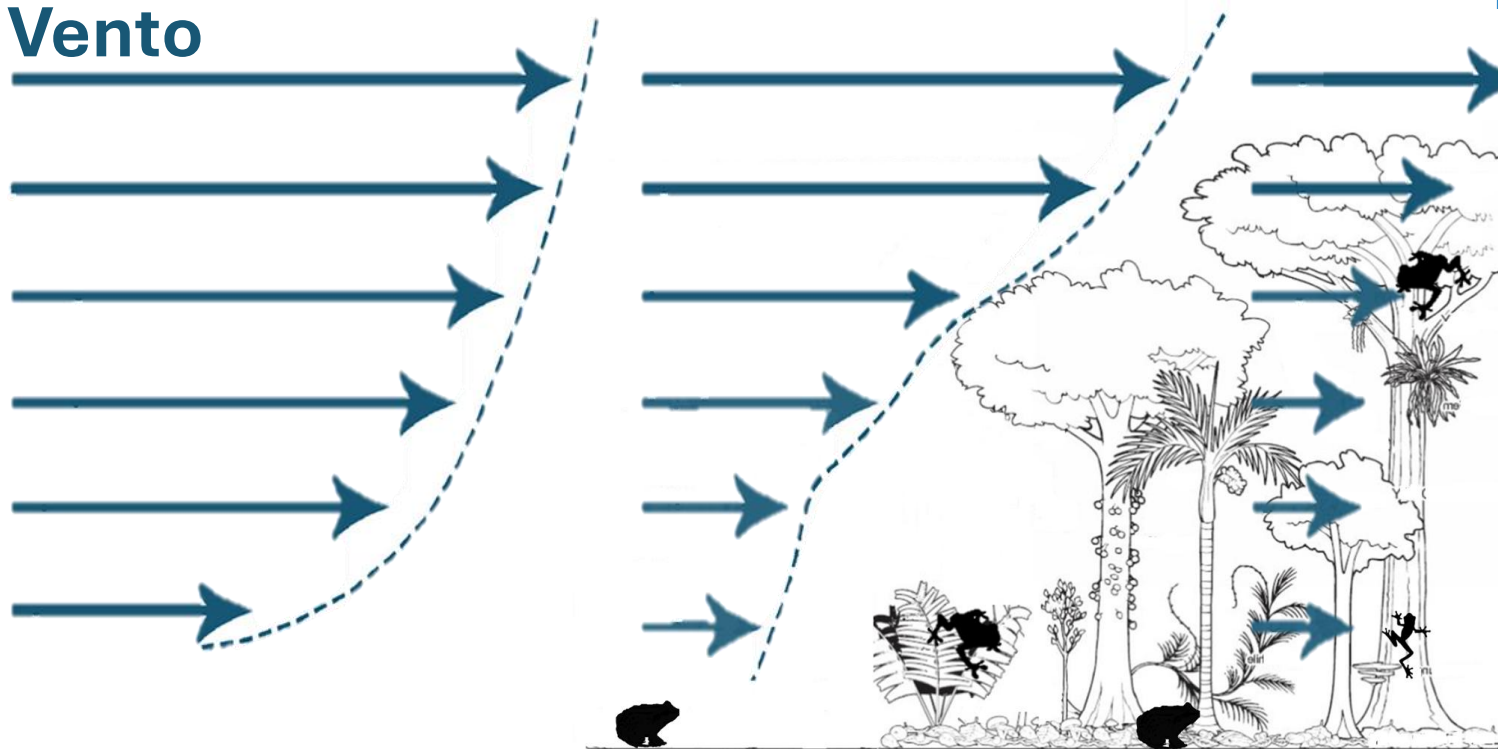
## MODELOS MACROCLIMÁTICOS



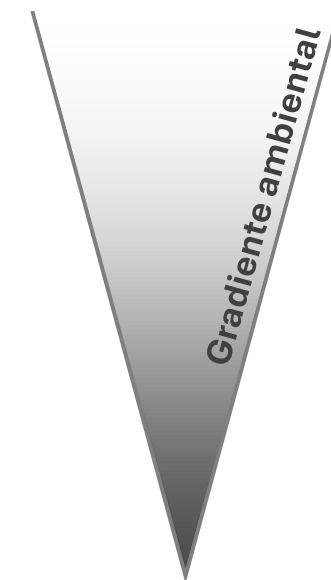
Microambiente



Vento

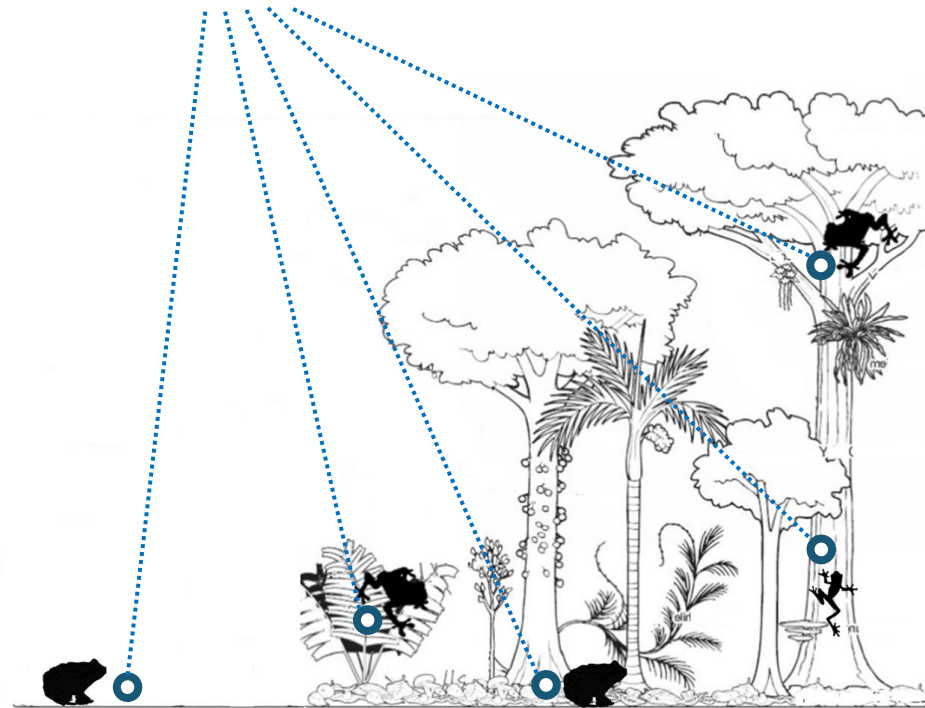


MODELOS MACROCLIMÁTICOS



Microambiente

## Microambiente





Macroambiente

Nivel de sombra

0%

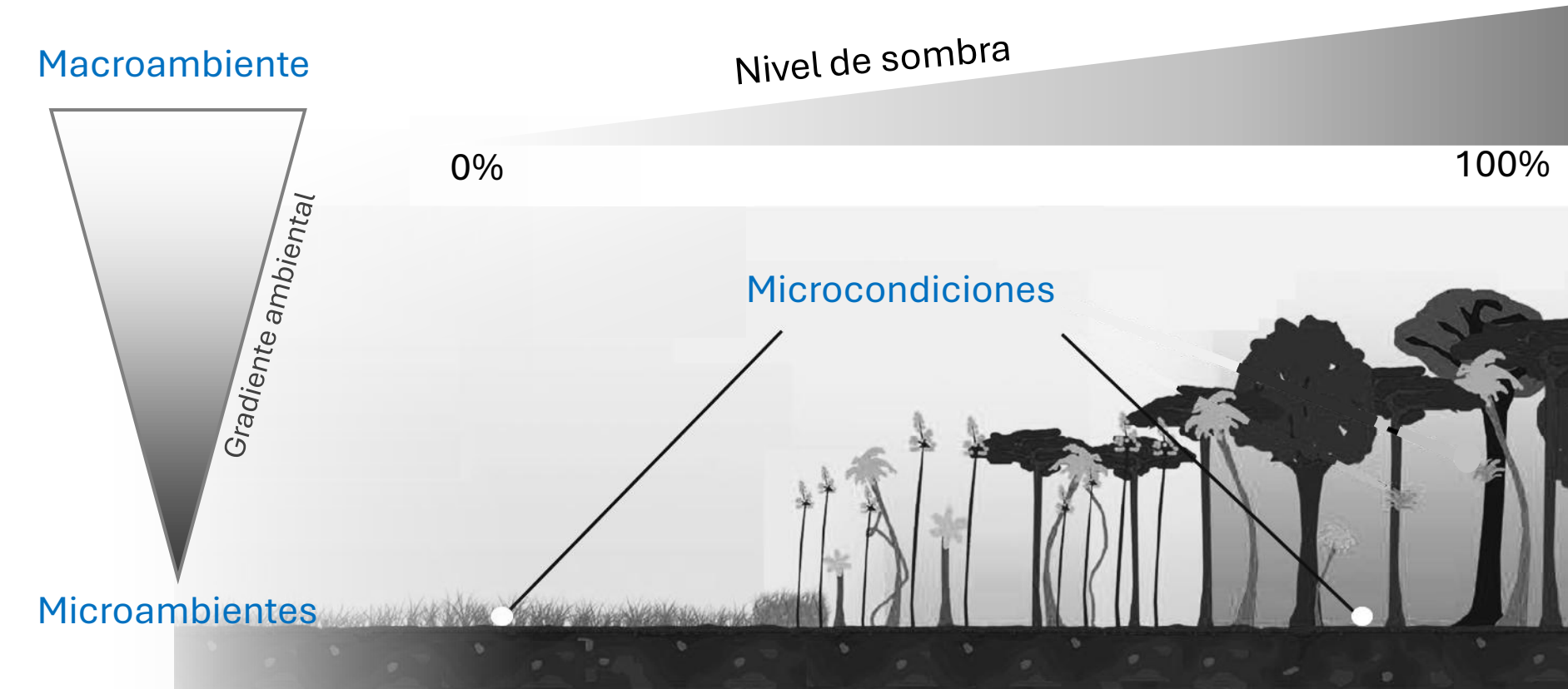
100%

Gradiente ambiental

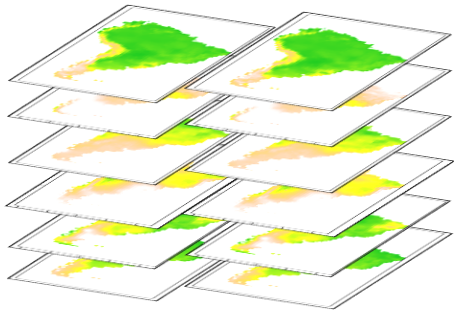
Microcondiciones

Microambientes

Paquete R **NicheMapR**



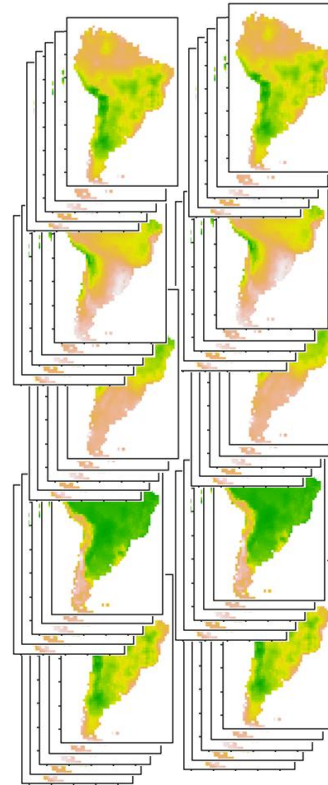
**Base climate dataset  
(New et al. 2002)**



Elevation  
Precipitation  
Wet-days  
Mean temperature  
Mean diurnal temperature range  
Mean relative humidity  
Mean wind speed  
Cloud cover

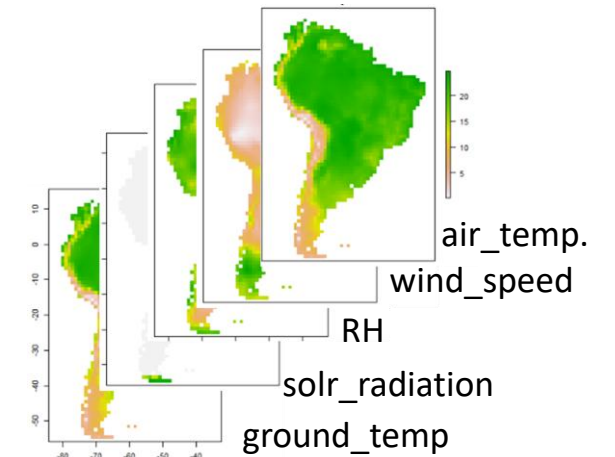


**97 raster  
maps  
(NicheMapR)**



global\_climate.nc  
soilw.mon.ltm.v2.nc

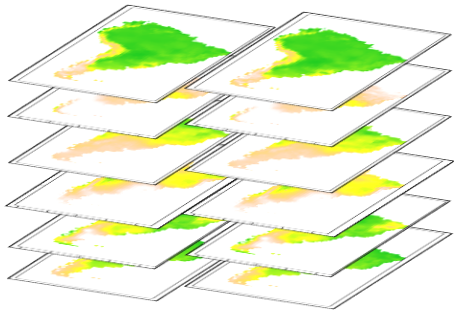
**Microclimatic variables**



## Departamento de Boyacá, Colombia



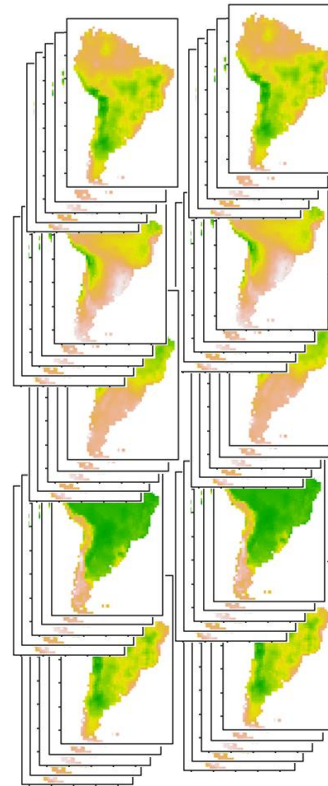
**Base climate dataset  
(New et al. 2002)**



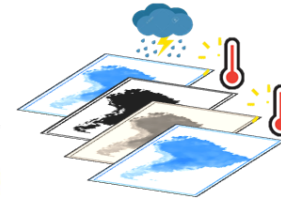
Elevation  
Precipitation  
Wet-days  
Mean temperature  
Mean diurnal temperature range  
Mean relative humidity  
Mean wind speed  
Cloud cover



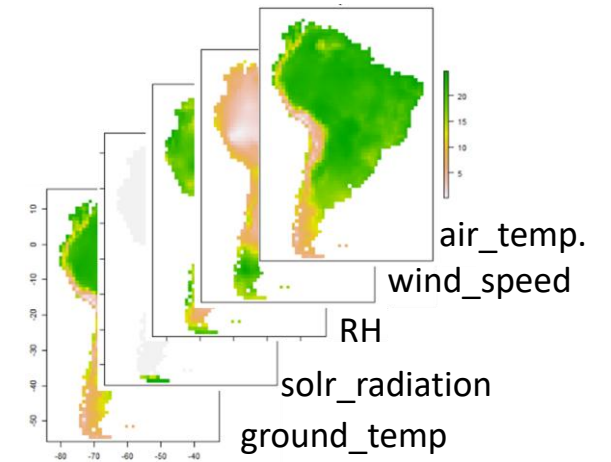
**97 raster  
maps  
(NicheMapR)**



[worldclim.org](http://worldclim.org)  
**CMIP6 - 2041-2060**  
GCM = ACCESS-CM2



**Microclimatic variables**



global\_climate.nc  
soilw.mon.ltm.v2.nc

